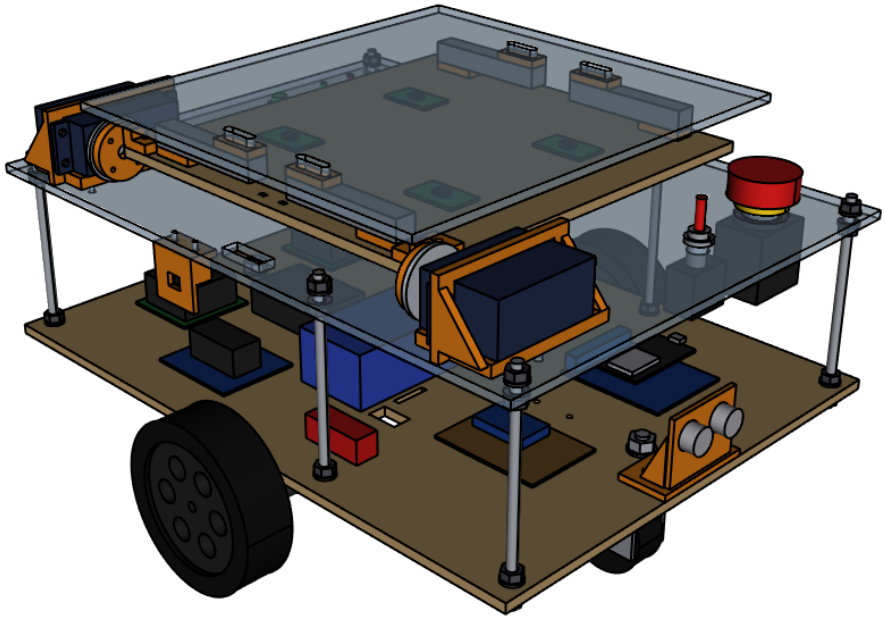
Cargo robot v2

Assembly instructions and list of components



Graduation Project II - 22 October 2026

Contents. 1 Components (bought, laser-cut, 3D-printed). 2 Tools. 3 Assembly, step by step, with a picture of every stage. 4 Final checks. Then drawing sheets 01 (general arrangement), 02 (layout of the electronics bay) and 07 (the tilting bed), at A3. The wiring is in CargoRobot_v2_connections.pdf.

All positions are in millimetres from the centre of the plates: +X is forward, +Y is left. The bed tips to the robot's RIGHT.

1. Components

1.1 Bought parts

No.	Group	Part	Qty	Make / buy
1	Compute	ESP32 DevKitC 38-pin (ESP32-WROOM-32, USB) + 38-pin screw-terminal breakout board	1	buy (get 2)
2	Compute	MCP23017 I2C 16-bit I/O expander module (address 0x20)	1	buy
3	Drive	JGB37-520 12V 170 RPM gear motor with Hall encoder (6 wires)	2	buy
4	Drive	Steel mounting bracket for 37 mm gear motor (+ its M3 screws)	2	buy
5	Drive	Wheel 100 mm x 30 mm with 6 mm D-shaft hub	2	buy
6	Drive	Swivel caster 50 mm, M8 threaded stem, >= 20 kg rating	2	buy
7	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
8	Sensors	8-channel IR line sensor array, TCRT5000, digital outputs, 3.3-5 V	1	buy
9	Sensors	Line sensor bracket (line_sensor_bracket.stl)	1	3D print PETG
10	Sensors	HC-SR04 ultrasonic distance sensor	1	buy (get 2)
11	Sensors	Ultrasonic sensor bracket (sonar_bracket.stl)	1	3D print PETG
12	Sensors	VL53L0X time-of-flight distance sensor module	1	buy
13	Sensors	Side sensor bracket (tof_bracket.stl)	1	3D print PETG
14	Payload	Straight-bar load cell 10 kg, 80x12.7x12.7 mm	4	buy
15	Payload	HX711 module	4	buy
16	Payload	Load-cell spacer (load_cell_spacer.stl)	8	3D print PETG 100 %
17	Payload	Cargo plate 260x280x5 (cargo_plate.dxf)	1	laser-cut acrylic
18	Tilting bed	Tilting plate 236x247x6 (tilt_plate.dxf)	1	laser-cut 6 mm plywood
19	Tilting bed	Servo 60 kg.cm, metal gears, 180 deg, 6-8.4 V, 25T spline, about 65x30x48 mm (e.g. DS5160)	2	buy
20	Tilting bed	25T aluminium disc servo horn, about 38 mm (often included with the servo)	2	buy
21	Tilting bed	Servo mount (servo_mount_front.stl + servo_mount_rear.stl)	2	3D print PETG
22	Tilting bed	Horn adapter (horn_adapter_front.stl + horn_adapter_rear.stl)	2	3D print PETG 100 %
23	Tilting bed	Bed rest post (bed_rest.stl) + 18 mm self-adhesive rubber pad 2 mm	2	3D print PETG
24	Tilting bed	Buck converter 10 A (e.g. 300 W XL4016 module), set to 7.4 V for the servos	1	buy
25	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
26	Power	XL4015 5 A buck converter (5.1 V for the ESP32 and sensors)	1	buy
27	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
28	Power	Emergency stop 22 mm panel-mount, latching, NC contact block	1	buy
29	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A	1	buy
30	Controls	Push button 12 mm panel-mount, momentary, normally open (start)	1	buy
31	Controls	5 mm LED green + red, with panel clips, + 330 ohm resistors	1	buy
32	Controls	Active buzzer module, 3-pin, 3.3-5 V	1	buy
33	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm plywood
34	Chassis	Top deck 400x300x5 (top_deck.dxf)	1	laser-cut 5 mm acrylic
35	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
36	Chassis	M6 nut + M6 washer	24	buy
37	Chassis	M8 nut + M8 washer (casters)	4	buy
38	Hardware	M4x12 screws + nylock nuts (motor brackets, servo mounts, rest posts), M4x16 + nuts (horn adapters), M4/M5x16 (load cells), M3 screws + nuts (sensor brackets), M2.5/M3 for the boards, VHB tape, velcro strap, zip ties	1	buy
39	Hardware	Resistors 1k x1, 2k x1, 10k x5, 2.2k x1, 4.7k x2, 100 nF x1, small perfboard x2	1	buy

1.2 Laser-cut plates (files in cad/output/dxf/)

File	Material	Size (mm)	Qty	Hole table
bottom_plate.dxf	6 mm birch plywood	400 x 300	1	sheet 03

File	Material	Size (mm)	Qty	Hole table
top_deck.dxf	5 mm clear acrylic	400 x 300	1	sheet 04
tilt_plate.dxf	6 mm birch plywood	236 x 247	1	sheet 05
cargo_plate.dxf	5 mm clear acrylic	260 x 280	1	sheet 06

1.3 3D-printed parts (files in cad/output/parts/; PETG, 0.2 mm layers, 4 walls)

File	Qty	Orientation	Infill	Supports
servo_mount_front.stl, servo_mount_rear.stl	1 + 1	foot flat on the bed	40 %	none
horn_adapter_front.stl, horn_adapter_rear.stl	1 + 1	round wall flat on the bed	100 %	on build plate only
bed_rest.stl	2	flat end down	40 %	none
load_cell_spacer.stl	8	flat	100 %	none
line_sensor_bracket.stl	1	flange flat on the bed	40 %	under the shelf
sonar_bracket.stl	1	base flat on the bed	40 %	none
tof_bracket.stl	1	top flange flat on the bed	40 %	none

Before printing the servo mounts, measure your servo's tab holes: the mounts assume 74 mm along the servo and 18 mm across (SERVO in cad/robot_cad.py).

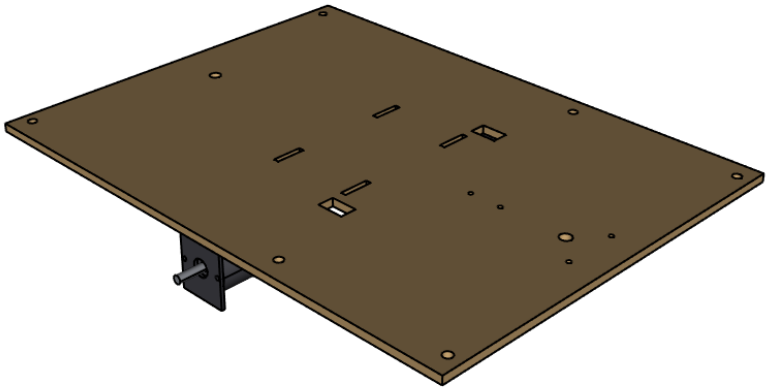
2. Tools

Soldering iron and solder; multimeter with a 10 A range; wire stripper, side cutters, ferrule crimper; hacksaw and file (threaded rod); drill with a 4.5 mm bit (motor brackets: the only holes you drill); spanners 7, 10 and 13 mm, hex keys, screwdrivers; heat gun; tape measure, square ruler; phone level / angle app; a 2 kg reference weight (rice weighed on a kitchen scale).

3. Assembly, step by step

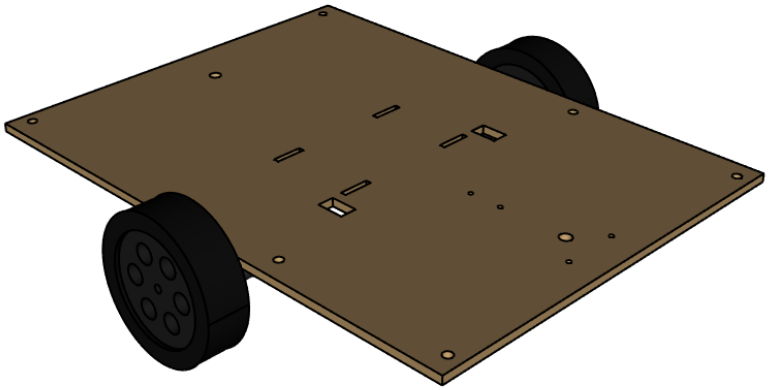
Use nylock nuts where listed, and don't over-tighten anything that touches acrylic: firm with a small spanner is enough. Each picture shows everything fitted so far.

Step 1: Motors under the bottom plate



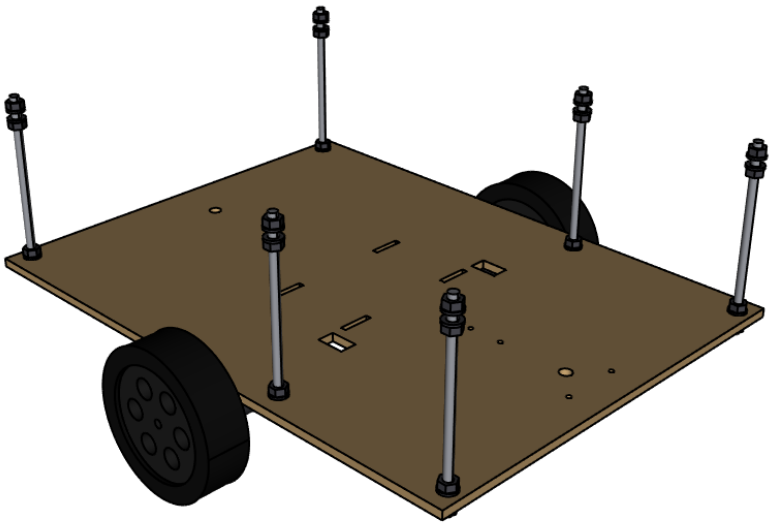
1. Fix each motor in its steel bracket with the bracket's M3 screws, shaft through the outer face.
2. Turn the bottom plate upside down. Place each bracket 5 mm in from the side edge, centred at X = 0, the motor pointing inwards. Mark its base holes, drill 4.5 mm, fix with M4 x 12 + washer + nylock nut.
3. Route each motor's 6-wire cable up through the nearest cable slot (X = +40, Y = +-70).

Step 2: Drive wheels



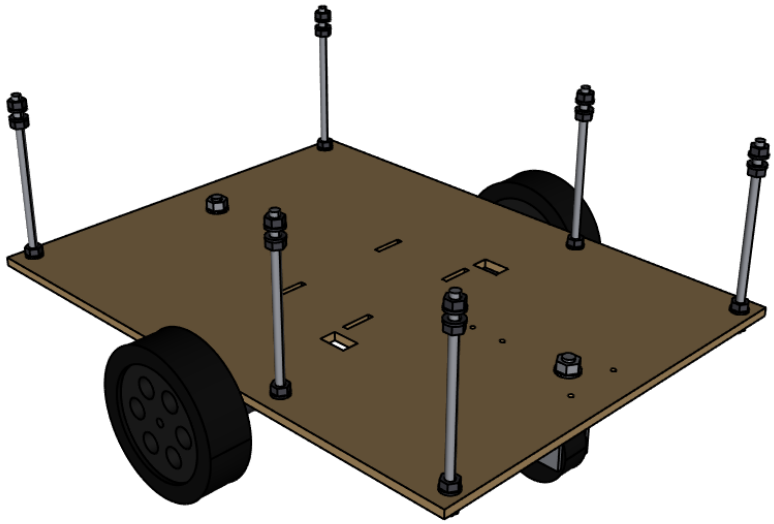
1. A drop of blue thread-locker on each set screw.
2. Slide the hub onto the 6 mm D-shaft, about 2 mm clear of the bracket; tighten the set screw on the flat.

Step 3: Threaded rods



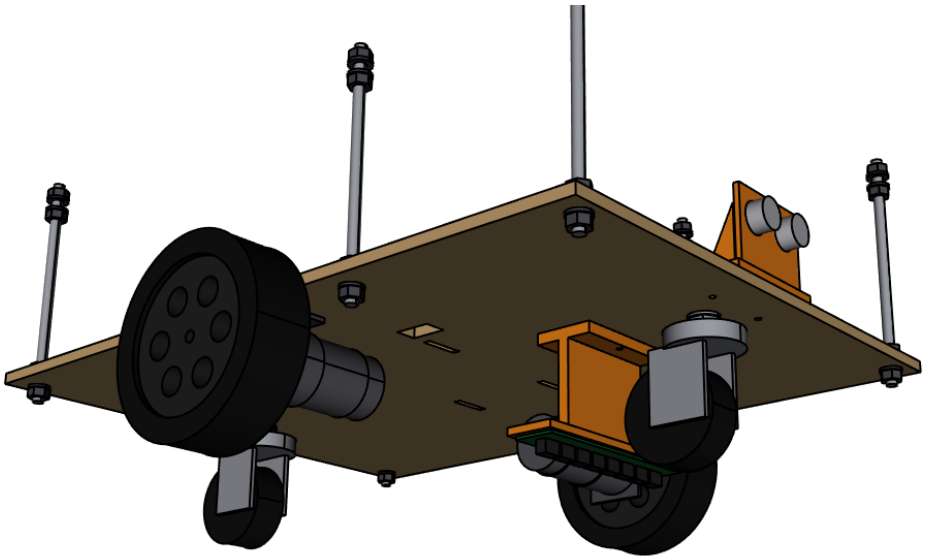
1. Cut 6 pieces of 130 mm from the M6 rod (screw a nut on first, unscrew it over the cut end).
2. Each rod: nut + washer 7 mm from the bottom end, up through the plate, washer + nut on top, tighten.
3. A third nut + washer on each rod with its top exactly 100 mm above the plate (use one gauge for all 6). The deck rests on these washers. Rods at X = -190, +60, +190, Y = +-135.

Step 4: Casters



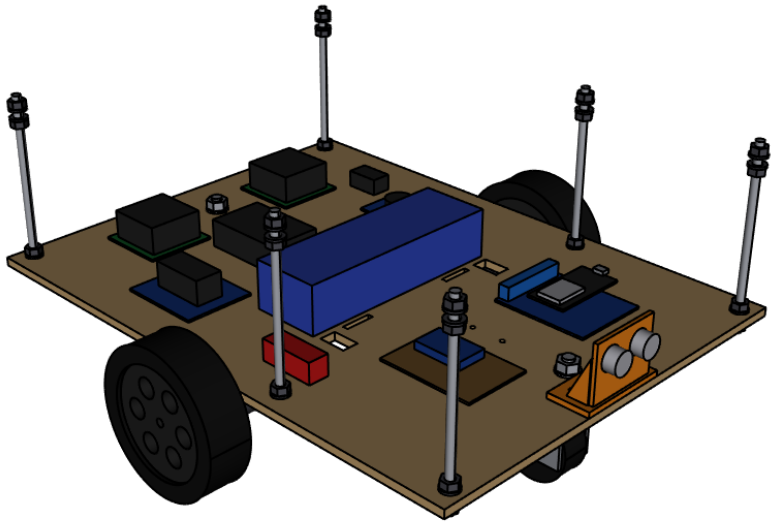
1. On each M8 stem: nut down to the caster body + washer, up through the 8.5 mm hole (X = +-160), washer + nut on top. The height is set later, with all 4 wheels on the floor.

Step 5: Sensors: line sensor underneath, ultrasonic in front



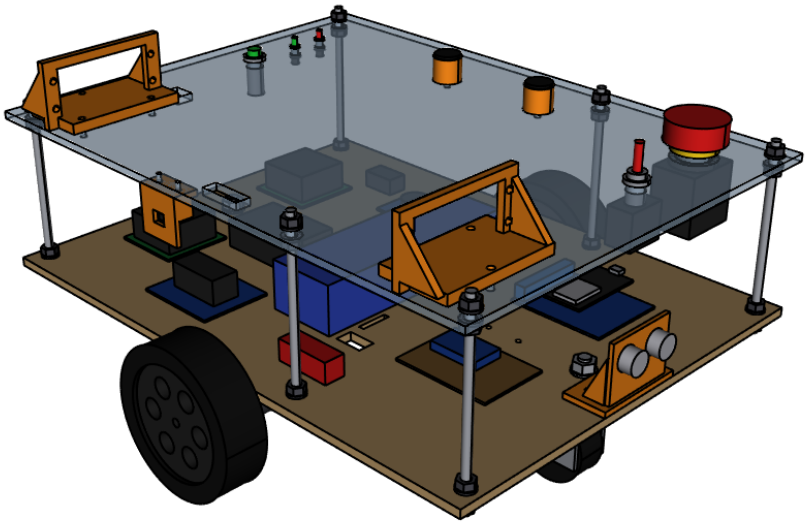
1. Screw the 8-channel line sensor under the shelf of the line-sensor bracket (2 x M3), sensors down, sensor 1 on the robot's LEFT.
2. Bolt the bracket under the bottom plate, between the drive wheels and the front caster: 2 x M3 x 12 through the 2 holes on the centre line (X = 98 +-12). The sensors end up about 5 mm above the floor.
3. Push the HC-SR04 into the ultrasonic bracket from behind (cans through the windows); hot glue or 2 zip ties. Bolt the bracket on top of the plate's front edge (2 x M3).
4. Check that the front caster still swivels a full turn without touching the line-sensor bracket.

Step 6: Electronics



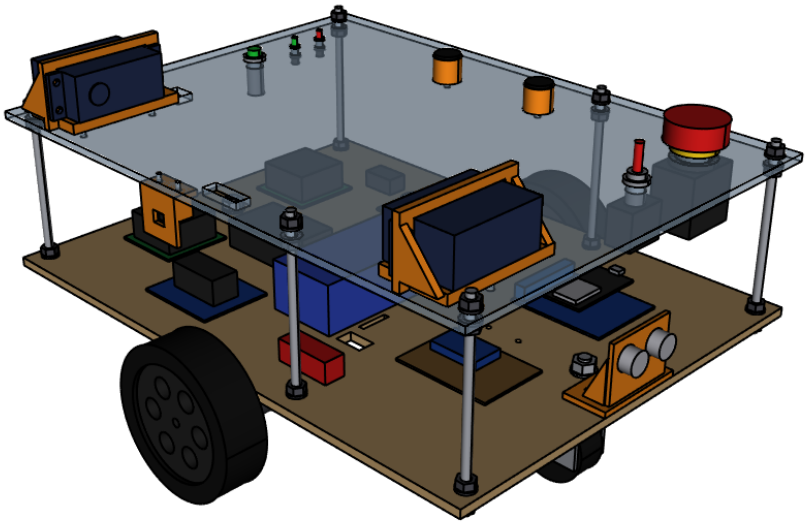
1. Stick every board down with VHB tape, where the picture shows (sheet 02 has the layout with numbers).
2. BTS7960 drivers at the rear, heatsinks up. ESP32 breakout at the front-left, USB socket towards the robot's left side. I/O board at the front-right. 5 V buck rear-left, 7.4 V servo buck rear-right, WAGO buses between the battery and the drivers.
3. The battery lies across the middle in 2 velcro straps (through the 4 slots) and slides out sideways.
4. Wiring: weeks 2 and 3 of BUILD_GUIDE.md and CargoRobot_v2_connections.pdf.

Step 7: Deck: controls, side sensor, servo mounts, bed rests



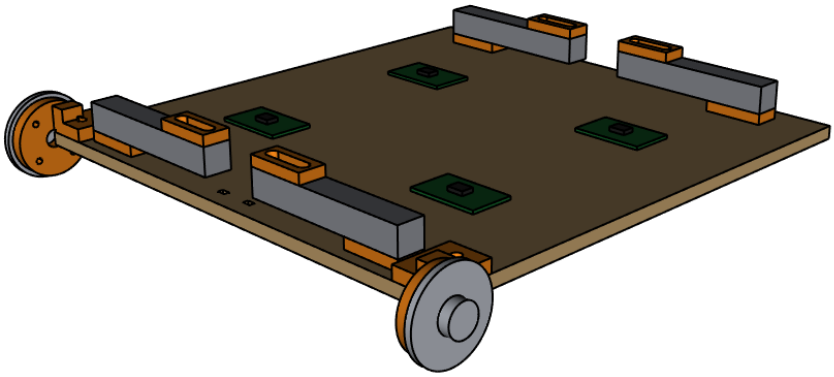
1. Before the deck goes on: E-stop through the 22.5 mm hole (front-left) from above, ring nut below, contact block clipped on, 2 x 40 cm 16 AWG on its NC terminals.
2. Master switch (front, middle) and start button (rear) up through their 12.2 mm holes; LEDs in their clips next to the button (green nearer the button).
3. Side sensor: VL53L0X behind the window of its bracket, bracket under the deck's right edge (2 x M3).
4. Servo mounts on the deck's front-right and rear-right corners (3 x M4 each); bed rests at X = +-40, Y = +95 (M4 from below into the nut in the post), rubber pad on top.
5. Lower the deck onto the 6 rods, washer + nut on top of each, check it is level.

Step 8: Servos



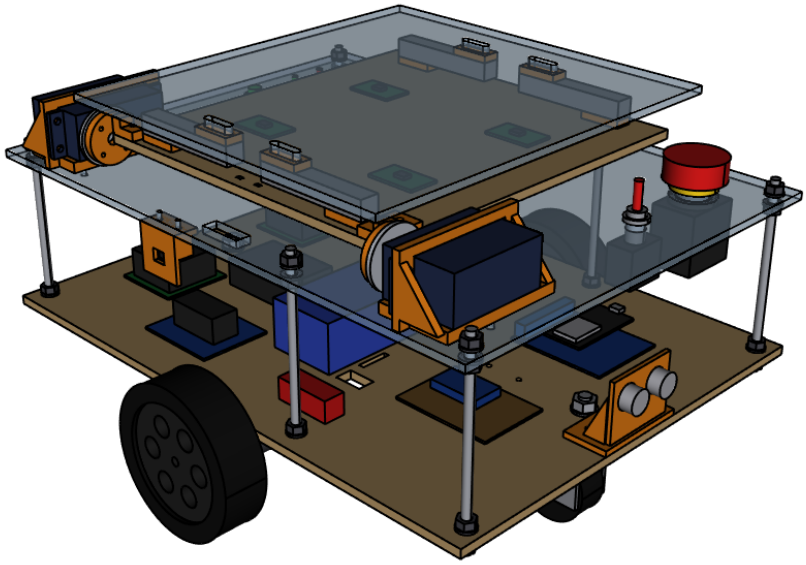
1. Each servo lies in its mount: body through the window, shaft towards the middle of the robot and at the deck's right edge, cable at the bottom. 4 x M4 x 12 + nut through the tabs.
2. Check your servo's tab holes against the mount first (74 x 18 mm in the model).
3. Servo cables down through the cut-outs next to the mounts.

Step 9: The bed (shown on its own)



1. Load cells on the tilting plate: spacer + cell's fixed end over each slot, arrow pointing down, 2 screws up from below into the cell. Fixed ends outwards, loaded ends towards the middle. The screws must not come out of the top of the cell.
2. HX711 boards on the plate between the cells (VHB tape).
3. Horn adapters on the plate's two right-hand corners (2 x M4 x 20 each, nuts underneath), then a disc horn on each adapter's round wall (4 x M3 x 10 into the 2.8 mm holes, thread-locker).

Step 10: Cargo plate on, bed on the servos



1. A spacer on each cell's loaded end, the cargo plate on top, 2 screws down through each slot into the cell. The cargo plate touches only the 4 spacers. No anti-slip mat: the package must slide off.
2. Lay the bed on its 2 rest posts and push the two horns onto the servo splines (turn a shaft gently by hand, one tooth at a time, until its horn fits). Fit the horns' centre screws with thread-locker.
3. Set the casters: all 4 wheels on the floor, the casters just touching (paper test).

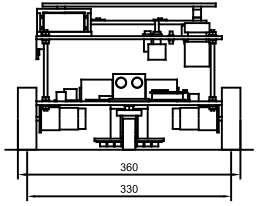
Step 11: Check: the bed swings freely



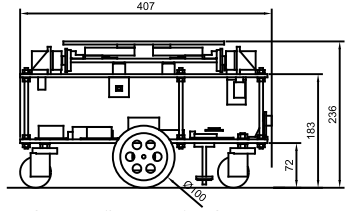
1. Lift the bed's left edge by hand (servos unpowered): it must swing to 45 deg and back without touching anything. The cargo plate's right edge ends just outside the deck.
2. The servo end points are set later with the 'servo' command (BUILD_GUIDE.md, week 4).

4. Final checks

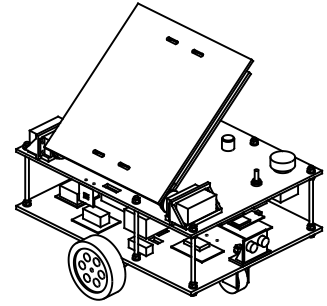
[ ] Push the robot by hand: it rolls straight, turns smoothly, the casters swivel freely, nothing rubs.
[ ] All 4 wheels on the floor: each caster grips a sheet of paper, but a firm tug still pulls it out.
[ ] 3 kg on the cargo plate: nothing bends visibly, the plate touches only the 4 spacers, no wheel lifts.
[ ] The bed swings from 0 to 45 deg by hand without touching anything, and lies level on its rests.
[ ] Press the E-stop hard: the deck doesn't bend noticeably.
[ ] Take photos of every stage for the report.



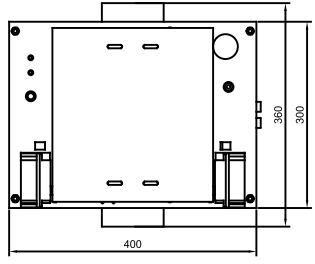
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW - BED TIPPED 40 deg

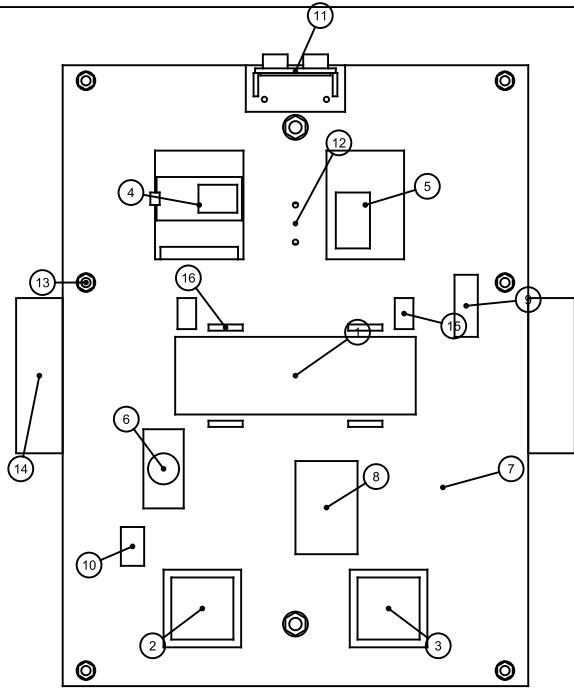


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 170 RPM gear motors, wheels D100 x 30.
3. Casters: 2 x 50 mm swivel, M8 threaded stem (height set by nuts).
4. Frame: 6 mm plywood + 5 mm acrylic on 6 x M6 threaded rod.
5. Bed: tipped 40 deg to the right by 2 servos on the hinge line.
6. Payload: 4 x 10 kg load cells under the cargo plate; packages up to 3 kg.
7. ESP32 + 8-ch IR line sensor (under), HC-SR04 (front), VL53L0X (right side).
8. Sheets: 01 GA, 02 layout, 03-06 plates, 07 bed, 08-09 prints, 10 BOM.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: GENERAL ARRANGEMENT					DWG No.: CR-V2-001
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:5	SHEET: 1 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:



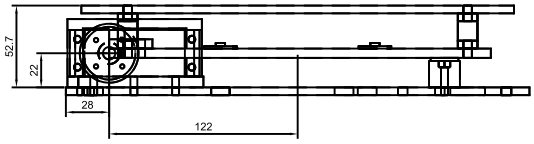
PLAN VIEW - DECK AND BED REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	ESP32 DevKit on a screw-terminal breakout (USB to the left)
5	I/O board: MCP23017 + resistor dividers
6	Buck converter 12 V -> 5.1 V (ESP32, sensors)
7	Buck converter 12 V -> 7.4 V (bed servos)
8	WAGO power buses (+12 V main / motor / 5 V / GND)
9	Inline fuse 10 A
10	Buzzer
11	HC-SR04 ultrasonic on printed bracket
12	8-channel IR line sensor (under the plate)
13	M6 threaded rod (x6)
14	Drive wheel D100 (motor under the plate)
15	Motor / encoder cable slot (x2)
16	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape.
Keep 10 cm slack on every wire that goes up to the deck or the bed.
Wiring: see BUILD_GUIDE.md weeks 2 and 3, and the schematics E01-E04.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LAYOUT - DECK REMOVED				DWG No.: CR-V2-002	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 2 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:



BED FLAT - FROM THE FRONT (front servo not shown)

HOW IT WORKS

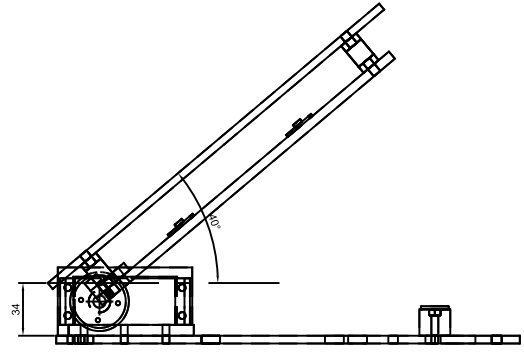
1. Two servos on the deck's right edge share one axis: their shafts ARE the hinge (front and rear end of the bed, mirror images). No hinge, screw or lever.
2. A printed horn adapter on each end of the tilting plate screws to the servo's aluminium disc horn (4 x M3). The load cells ride on the tilting plate.
3. Flat, the bed rests on 2 printed posts and the servos are switched off.
4. To unload, the ESP32 turns both servos slowly (20 deg/s) to 40 deg, waits until the load cells read empty, then lowers the bed and switches them off.
5. No anti-slip mat: a cardboard box slides on acrylic from about 20 deg.

TORQUE AT THE START OF THE TIP (both servos, 20 cm box)

PACKAGE	TORQUE	OF 2 x 60 kg.cm
1 kg	2.3 N.m	20 %
3 kg	4.7 N.m	40 %
5 kg	7.1 N.m	60 %

SERVOS

Servo: 60 kg.cm at 7.4 V, metal gears, 180 deg, 25T spline,
about 65 x 30 x 48 mm (e.g. DS5160). Check its tab holes:
74 x 18 mm here (SERVO in robot_cad.py).
Supply: 7.4 V from its own 10 A buck converter, after the E-stop.
Signal: front servo GPIO 2, rear servo GPIO 12 (schematic E03).
Hinge axis: 205 mm above the floor, 122 mm right of the centre line.



BED TIPPED 40 deg - THE PACKAGE SLIDES OFF TO THE RIGHT

(the right edge of the cargo plate ends up 6 mm outside the deck, 34 mm above it)

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING BED - SERVO HINGE				DWG No.: CR-V2-007	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 7 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN: